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SEPARATOR FOR ELECTROCHEMICAL DEVICE AND ELECTROCHEMICAL DEVICE INCLUDING THE SAME

Abstract

A separator for an electrochemical device includes a fluid inlet; a fluid outlet; a plurality of serpentine-type flow paths disposed between the fluid inlet and the fluid outlet; and an auxiliary flow path extending in a first direction and connected to inlets and outlets of the plurality of serpentine-type flow paths, the first direction being a direction from the fluid inlet toward the fluid outlet. The plurality of serpentine-type flow paths include a first serpentine-type flow path and a second serpentine-type flow path, and an inlet and an outlet of the second serpentine-type flow path are disposed closer to the fluid outlet in the first direction, compared to an inlet and an outlet of the first serpentine-type flow path.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S)

[0001] This application claims benefit of priority to Korean Patent Application No. 10-2024-0027154 filed on Feb. 26, 2024 in the Korean Intellectual Property Office, the disclosure of which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

[0002] The present disclosure relates to a separator for an electrochemical device, and the electrochemical device.

[0003] An electrochemical device may include a fuel cell generating electrical energy by electrochemically reacting a fuel (hydrogen) and an oxidizing agent (pure oxygen or atmospheric oxygen), an electrolysis cell generating hydrogen and oxygen through electrolysis of water, and the like.

[0004] As an example of such an electrochemical device, a solid oxide fuel cell (SOFC) and a solid oxide electrolysis cell (SOEC) may include a cell comprising an air electrode, a fuel electrode, and a solid electrolyte with oxygen ion conductivity, and the cell may be referred to as a solid oxide cell. The solid oxide cell may produce electrical energy through an electrochemical reaction or may produce hydrogen by electrolyzing water through a reverse reaction of a solid oxide fuel cell. In addition, other types of fuel cells or water electrolytic cells, such as a phosphoric acid fuel cell (PAFC), an alkaline fuel cell (AFC), a polymer electrolyte fuel cell (PEMFC), and a direct methanol fuel cell (DMFC), are also used in one form of the electrochemical device.

[0005] For the electrochemical device, it is common to use a stack structure in which unit cells are disposed between a pair of separators. Here, a flow path through which fluid may flow is formed in the separator. Water vapor, hydrogen, oxygen gas, and the like, may flow through the flow path of the separator, and a direction, a speed, a flow rate, or the like of the fluid may have a significant impact on performance of electrochemical devices. Accordingly, research has recently been conducted in the field of technology to optimize a size, a shape, or the like of the flow path.

SUMMARY

[0006] An aspect of the present disclosure is to implement a separator for an electrochemical device designed to provide high performance when applied to the electrochemical device.

[0007] As a method to solve the above-described problems, the present disclosure proposes a novel structure of a separator for an electrochemical device through an example embodiment, and specifically, the separator includes a fluid inlet; a fluid outlet; a plurality of serpentine-type flow paths disposed between the fluid inlet and the fluid outlet; and an auxiliary flow path extending in a first direction and connected to inlets and outlets of the plurality of serpentine-type flow paths, the first direction being a direction from the fluid inlet toward the fluid outlet. The plurality of serpentine-type flow paths include a first serpentine-type flow path and a second serpentine-type flow path, and an inlet and an outlet of the second serpentine-type flow path are disposed closer to the fluid outlet in the first direction, compared to an inlet and an outlet of the first serpentine-type flow path.

[0008] In an embodiment, the auxiliary flow path may include a straight flow path.

[0009] In an embodiment, the auxiliary flow path may be parallel to the first direction.

[0010] In an embodiment, the auxiliary flow path may be connected to the fluid inlet and the fluid outlet, to have an integral structure.

[0011] In an embodiment, the first serpentine-type flow path may be provided as a pair on one side and the other side of the auxiliary flow path.

[0012] In an embodiment, an inlet of a serpentine-type flow path disposed closest to the fluid inlet, among the plurality of serpentine-type flow paths, may be connected to the fluid inlet.

[0013] In an embodiment, an outlet of a serpentine-type flow path disposed closest to the fluid outlet, among the plurality of serpentine-type flow paths, may be connected to the fluid outlet.

[0014] In an embodiment, the auxiliary flow path may be disposed in a central region in a second direction perpendicular to the first direction.

[0015] In an embodiment, a symmetrical structure may be formed, based on the auxiliary flow path.

[0016] In an embodiment, the plurality of serpentine-type flow paths may include a first flow path in the first direction and a second flow path in a second direction perpendicular to the first direction.

[0017] In an embodiment, the second flow path may be provided as a plurality of second flow paths, the first flow path may be connected to the plurality of second flow paths, and the plurality of second flow paths may be connected to the auxiliary flow path.

[0018] In an embodiment, the auxiliary flow path may be provided as a plurality of auxiliary flow paths, and the fluid inlet and the fluid outlet may be provided as a plurality of fluid inlets and as a plurality of fluid outlets, respectively, to be connected to the auxiliary flow path.

[0019] In an embodiment, the plurality of auxiliary flow paths may be disposed in an outer peripheral region in a second direction perpendicular to the first direction.

[0020] In an embodiment, the plurality of serpentine-type flow paths may include a first flow path in the first direction and a plurality of second flow paths in the second direction, the first flow path may be connected to the plurality of second flow paths, and the plurality of second flow paths may be connected to the plurality of auxiliary flow paths.

[0021] In an embodiment, the first flow path may be disposed in a central region in the second direction.

[0022] In an embodiment, a symmetrical structure may be formed, based on the first flow path.

[0023] A width, in a second direction perpendicular to the first direction, of the auxiliary flow path is less than a width, in the first direction, of each of the first serpentine-type flow path and the second serpentine-type flow path.

[0024] According to another aspect of the present disclosure, an electrochemical device includes a plurality of separators; and an electrochemical cell disposed between the plurality of separators. At least one of the plurality of separators includes a fluid inlet; a fluid outlet; a plurality of serpentine-type flow paths disposed between the fluid inlet and the fluid outlet; and an auxiliary flow path extending in a first direction and connected to inlets and outlets of the plurality of serpentine-type flow paths, the first direction being a direction from the fluid inlet toward the fluid outlet. The plurality of serpentine-type flow paths include a first serpentine-type flow path and a second serpentine-type flow path, and an inlet and an outlet of the second serpentine-type flow path are disposed closer to the fluid outlet in the first direction, compared to an inlet and an outlet of the first serpentine-type flow path.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0025] The above and other aspects, features, and advantages of the present disclosure will be more clearly understood from the following detailed description, taken in conjunction with the accompanying drawings, in which:

[0026] FIGS. 1 and 2 schematically illustrate an appearance of a separator for an electrochemical

device according to an embodiment of the present disclosure, and correspond to a perspective view and a plan view, respectively.

[0027] FIG. 3 illustrates flow of fluid in the separator of FIGS. 1 and 2.

[0028] FIG. 4 is a plan view schematically illustrating an appearance of a separator for an electrochemical device according to a modified embodiment.

[0029] FIG. 5 illustrates flow of fluid in the separator of FIG. 4.

[0030] FIGS. 6 to 9 illustrate a distribution of a mole fraction of hydrogen and a flow velocity in a separator according to a conventional technology.

[0031] FIGS. 10 to 13 illustrate a distribution of a mole fraction of hydrogen and a flow velocity in a separator according to a separator according to an embodiment of the present disclosure.

[0032] FIG. 14 is a graph comparing a hydrogen production rate according to a shape of a separator.

[0033] FIG. 15 illustrates a water electrolysis device to which a separator according to an embodiment of the present disclosure is applied.

DETAILED DESCRIPTION

[0034] Hereinafter, embodiments of the present disclosure will be described with reference to specific embodiments and the attached drawings. The embodiments of the present disclosure may, however, be exemplified in many different forms and should not be construed as being limited to the specific embodiments set forth herein. Furthermore, the embodiments disclosed herein are provided for those skilled in the art to better explain the present disclosure. Accordingly, in the drawings, the shapes and dimensions of elements may be exaggerated for clarity, and the same reference numerals will be used throughout to designate the same or like elements.

[0035] Furthermore, in order to clearly describe the present disclosure in the drawings, contents unrelated to the description are omitted, and since sizes and thicknesses of each component illustrated in the drawings are arbitrarily illustrated for convenience of description, the present disclosure is not limited thereto. Furthermore, components with the same function within the same range of ideas are described using the same reference numerals. Throughout the specification, when a certain portion “includes” or “comprises” a certain component, this indicates that other components are not excluded and may be further included unless otherwise noted.

[0036] FIGS. 1 and 2 schematically illustrate an appearance of a separator for an electrochemical device according to an embodiment of the present disclosure, and correspond to a perspective view and a plan view, respectively. FIG. 3 illustrates flow of fluid in the separator of FIGS. 1 and 2.

[0037] Referring to FIGS. 1 and 2, a separator **100** for an electrochemical device according to an embodiment of the present disclosure (hereinafter, referred to as a “separator for an electrochemical device” or a “separator”) may include a fluid inlet **111**, a fluid outlet **112**, a serpentine-type flow path **120**, and an auxiliary flow path **130**. The serpentine-type flow path **120** and the auxiliary flow path **130** may be disposed between the fluid inlet **111** and the fluid outlet **112**. In addition, in the serpentine-type flow path **120**, an inlet I and an outlet O may be connected to the auxiliary flow path **130**, and a second serpentine-type flow path **120b** may be disposed relatively closer to the fluid outlet **112** than a first serpentine-type flow path **120a**. For example, if a direction from the fluid inlet **111** toward the fluid outlet **112** is referred to as a first direction **D1**, the inlet I and the outlet O of the second serpentine-type flow path **120b** may be disposed closer to the fluid outlet **112**, as compared to an inlet I and an outlet O of the first serpentine-type flow path **120a**, in the first direction **D1**. With this type of flow path structure, the separator **100** may have high efficiency when applied to an electrochemical device. For example, when the separator **100** is used in a water electrolysis device, since fuel (e.g., water vapor) may be supplied from the auxiliary flow path **130** to the serpentine-type flow path **120**, production efficiency of hydrogen may increase in an overall region of the separator **100**. In addition, even when the separator **100** is used in a fuel cell device, since fuel (e.g., hydrogen) may be supplied from the auxiliary flow path **130** to the serpentine path **120**, production efficiency of power may increase in an overall region of the separator **100**. More

detailed descriptions of technical features of the auxiliary flow path **130** and the serpentine path **120** will be described later.

[0038] The separator **100** may be a component separating unit cells in the electrochemical device, and may be formed of a conductive material. Specifically, when the separator **100** is used as an SOEC or an SOFC, the separator **100** may include a metal having a high melting point, to not melt or soften at high temperatures. For example, the separator **100** may use a nickel-based material, an iron-based material, a stainless-based material, or the like. In addition, when an operating temperature of the separator **100** is relatively low, for example, at about 800° C. or lower, copper, a copper alloy, or the like having good conductivity may also be used. The separator **100** may include a wall **110** to form a flow path, and may include the auxiliary flow path **130** and the serpentine-type flow path **120**, as illustrated. In this case, the fluid inlet **111** and the fluid outlet **112** of the separator **100** do not need to be clearly distinguished from the flow paths **120** and **130**, and the fluid inlet **111** and the fluid outlet **112** may not be provided as separate components, but may form a portion of the flow paths **120** and **130**. To form the wall **110** in the separator **100**, for example, a metal plate may be processed using a stamping process, an etching process, or the like, as an appropriate process. When the separator **100** includes the fluid inlet **111** and the fluid outlet **112**, and is used, for example, as a water electrolysis device, water may be injected as water vapor through the fluid inlet **111** and hydrogen gas may be discharged through the fluid outlet **112**, and when the separator **100** is used as a fuel cell, the separator **100** may be operated in an opposite manner to what was described above. The fluid inlet **111** and the fluid outlet **112** may be changed, depending on a direction of fluid injection, and accordingly, the inlet I and the outlet O of the serpentine-type flow path **120** may also be changed. Hereinafter, a function of the wall **110** will be described by taking as a main example in a case in which the separator **100** is applied to a fuel electrode of the water electrolysis device. In a case in which the separator **100** is disposed on the fuel electrode of the water electrolysis device, water vapor and a small amount of hydrogen may be injected from the fluid inlet **111**, and as the fluid flows along the separator **100**, a concentration of the water vapor may decrease and a concentration of the hydrogen may increase. The flow paths of the separator **100** may have functions of supplying reactants while moving a product generated by a reaction therebetween to the fluid outlet **112** to effectively remove the same.

[0039] As described above, the separator **100** may include the serpentine-type flow path **120** and the auxiliary flow path **130** connected to the inlet I and the outlet O of the serpentine-type flow path **120**. In this case, a boundary between the serpentine-type flow path **120** and the auxiliary flow path **130** may not be clearly distinguished, and the serpentine-type flow path **120** and the auxiliary flow path **130** may be connected to each other to form an integral structure. The serpentine-type flow path **120** may correspond to a main flow path as a curved flow path. In terms of shape, the serpentine-type flow path **120** may include a section in which a fluid path changes at least once. As a specific example, a plurality of serpentine-type flow paths **120** may include a first flow path **121** in the first direction D1 and a second flow paths **122** in a second direction D2. In this case, the second flow path **122** may be provided as a plurality of second flow paths **122**, the first flow path **121** may be connected to the plurality of second flow paths **122**, and the plurality of second flow paths **122** may be connected to the auxiliary flow paths **130**. Although the present embodiment illustrates the serpentine-type flow path **120** in which a fluid path changes twice, the number of sections of changing the fluid path may be 1 or 3 or more. In addition, the serpentine-type flow path **120** in the present embodiment has a region bent at a right angle, but may also have a region bent at a non-right angle or bent to form a curve.

[0040] The auxiliary flow path **130** may extend in the first direction D1 as a whole, and may include a straight flow path. Therefore, the auxiliary flow path **130** may induce a fluid path in the first direction D1. The auxiliary flow path **130** does not have to have a single straight shape in an entire region, and there may be a region that deviates from a single straight line in some sections. As a specific example, as illustrated in FIG. 2, the auxiliary flow path **130** may be parallel to the

first direction **D1**. In addition, the auxiliary flow path **130** may be connected to the fluid inlet **111** and the fluid outlet **112** to form an integral structure. As described above, the first direction **D1** may be defined as a direction from the fluid inlet **111** to the fluid outlet **112** in the separator **100**, but the first direction **D1** does not represent only one direction, and may include both directions, e.g., upward and downward directions with respect to FIG. 2. Likewise, the second direction **D2**, perpendicular to the first direction **D1**, does not represent only one direction, but may include both directions, e.g., left and right directions, based on FIG. 2. Likewise, a third direction **D3**, perpendicular to the first and second directions **D1** and **D2**, does not represent only one direction, but may include both directions, e.g., two directions that may be opposite to each other.

[0041] In the present embodiment, the serpentine-type flow path **120** may be provided as a plurality of serpentine-type flow paths **120**. Specifically, the serpentine-type flow path **120** may include the first and second serpentine-type flow paths **120a** and **120b**, which may be conveniently named according to the order in which they are disposed based on the fluid inlet **111** and the fluid outlet **112**. For example, when naming each of the plurality of serpentine-type flow paths **120**, they may be named in order from the closest to the fluid inlet **111**, such as the first serpentine-type flow path **120a** and the second serpentine-type flow path **120b**. In this case, the serpentine-type flow path disposed closest to the fluid inlet **111** among the plurality of serpentine-type flow paths **120**, i.e., the first serpentine-type flow path **120a** may have the inlet I connected to the fluid inlet **111**. In addition, the serpentine-type flow path disposed closest to the fluid outlet **112** among the plurality of serpentine-type flow paths **120**, i.e., a serpentine-type flow path disposed in an uppermost portion with reference to FIG. 2 may have an outlet O connected to the fluid outlet **112**.

[0042] As in the present embodiment, while employing the serpentine-type flow path **120** and the auxiliary flow path **130** in the separator **100**, and optimizing arrangement of the serpentine-type flow paths **120**, e.g., arranging them in the order of the inlet I and the outlet O of the first serpentine-type flow path **120a**, and the inlet I and the outlet O of the second serpentine-type flow path **120b**, fuel such as water vapor may be effectively supplied to the serpentine-type flow path **120** through the auxiliary flow path **130** from the fluid inlet **111** to the fluid outlet **112**. By re-supply of the fuel, production efficiency of hydrogen may be improved even in a region adjacent to the fluid outlet **111**. With reference to FIGS. 2 and 3 in detail, the serpentine-type flow path **120** may correspond to the main flow path, and fuel such as water vapor or the like may exhibit a flow indicated by a thick dotted line. In using only a straight flow path or a serpentine-type flow path like conventional separators, even when both the straight flow path and the serpentine-type flow path are used, and when an arrangement method as in the present embodiment is not used, there may be a problem that production efficiency of hydrogen decreases because a molar fraction of hydrogen as a product continuously increases from a low level as fuel is consumed as the fuel passes through the flow paths. In the present embodiment, to solve the problem, the fuel may be designed to be resupplied from the auxiliary flow path **130** to the serpentine-type flow path **120**, more specifically, to the outlet O of the serpentine-type flow path **120** and a region adjacent thereto. Therefore, the mole fraction of hydrogen in the serpentine-type flow path **120** may discontinuously increase, and production efficiency of hydrogen may be improved. In a fuel cell device, a molar fraction of water in the serpentine-type flow path **120** may be discontinuously increased, and production efficiency of power may be improved.

[0043] To explain more specific examples of shapes and an arrangement of flow paths of the separator **100**, first, the first serpentine-type flow path **120a** may be provided as a pair of first serpentine-type flow paths **120a** on one side and the other side of the auxiliary flow path **130**, respectively. For example, the first serpentine-type flow path **120a** may be provided as a pair of first serpentine-type flow paths **120a** connected to the auxiliary flow path **130**, and may be disposed on left and right sides of the auxiliary flow path **130**, respectively, based on FIG. 2. In addition, the second serpentine-type flow path **120b** or other additional serpentine-type flow paths may also have the same symmetrical structure. In this case, as illustrated, the auxiliary flow path

130 may be disposed in a central region of the separator **100** in the second direction **D2**. In addition, the separator **100** may have a symmetrical structure as a whole, centered on the auxiliary flow path **130**. In one example, a width **W1**, in the second direction **D2**, of the auxiliary flow path **130** may be less than a width **W2**, in the first direction **D1**, of each of the first serpentine-type flow path **120a** and the second serpentine-type flow path **120b**.

[0044] In this manner, in the present embodiment, a separator **100** in which fuel may be replenished from the auxiliary flow path **130** to the serpentine-type flow path **120** corresponding to a main path may be used to improve reactivity of the fuel and production efficiency of products (hydrogen, power, or the like). In addition to the structure illustrated in the above-described embodiment, the serpentine-type flow path **120** and the auxiliary flow path **130** may be disposed in other manners, which will be described with reference to FIGS. 4 and 5. FIG. 4 may be a plan view schematically illustrating an appearance of a separator for an electrochemical device according to a modified embodiment, corresponding to FIG. 2. FIG. 5 illustrates flow of fluid in the separator of FIG. 4, corresponding to FIG. 3. As in the present modified example, an auxiliary flow path **130** may be provided as a plurality of auxiliary flow paths **130**. In addition, a fluid inlet **111** and a fluid outlet **112** may be provided as a plurality of fluid inlets **111** and a plurality of fluid outlets **112**, respectively, to correspond to the plurality of auxiliary flow paths **130**, and furthermore, the fluid inlet **111** and the fluid outlet **112** may be connected to the auxiliary flow path **130**. When the plurality of auxiliary flow paths **130** are provided, the plurality of auxiliary flow paths **130** may be disposed in an outer peripheral region in the second direction **D2**. In addition, a plurality of serpentine-type flow paths **120a** and **120b** may include a first flow path **121** in the first direction **D1** and a second flow path **122** in the second direction **D2**. In this case, the second flow path **122** may be provided as a plurality of second flow paths **122**, the first flow path **121** may be connected to the plurality of second flow paths **122**, and the plurality of second flow paths **122** may be connected to the plurality of auxiliary flow paths **130**. When the plurality of auxiliary flow paths **130** are disposed in the outer peripheral region in the second direction **D2**, the first flow path **121** of the serpentine-type flow paths **120a** and **120b** may be disposed in a central region in the second direction **D2**. In this case, the separator may have a symmetrical structure as a whole, centered on the first flow path **121**.

[0045] In the present modified example, fuel may be replenished from the auxiliary flow path **130** disposed on an outside to the serpentine-type flow path **120** corresponding to a main path, and reactivity of the fuel and production efficiency of products (hydrogen, power, or the like) may be improved therefrom.

[0046] Referring to FIGS. 6 to 14, results of comparing characteristics according to a shape of a flow path of a separator may be described. First, FIGS. 6 to 13 illustrate a distribution of a mole fraction of hydrogen and a flow velocity in a separator according to a conventional technology and a separator according to an embodiment of the present disclosure. In this case, FIG. 6 (hydrogen mole fraction) and FIG. 7 (flow velocity) may be for a separator that only applies a straight flow path, and FIG. 8 (hydrogen mole fraction) and FIG. 9 (flow velocity) may be for a separator that only applies a serpentine-type flow path. FIG. 10 (hydrogen mole fraction) and FIG. 11 (flow velocity) may be for a separator according to the embodiment of FIG. 2, and FIG. 12 (hydrogen mole fraction) and FIG. 13 (flow velocity) may be for a separator according to the embodiment of FIG. 4. In addition, FIG. 14 may be a graph comparing hydrogen production rates according to a shape of a separator, and {circle around (1)} is a result for a separator including only a straight flow path, {circle around (2)} is a result for a separator including only a serpentine-type flow path, {circle around (3)} is a result for a separator according to the embodiment of FIG. 2, and {circle around (4)} is a result for a separator according to the embodiment of FIG. 4.

[0047] According to simulation results of FIGS. 6 to 13, in a separator having a composite flow path structure according to an embodiment of the present disclosure, as compared to a separator using only a straight flow path or only a serpentine-type flow path, a mole fraction of hydrogen,

which had been increased as fuel was replenished in a region in which the straight flow path and the serpentine-type flow path meet, showed a tendency to decrease again, and accordingly, as can be seen in the graph of FIG. 14, production efficiency of hydrogen by chemical reaction was significantly improved, as compared to the conventional separator.

[0048] An example of applying the separator described above to an electrochemical device will be described with reference to FIG. 15. FIG. 15 illustrates a water electrolysis device to which a separator having a linear wall is applied. The separator described above may be applied to other types of electrochemical devices such as a fuel cell, not the water electrolysis devices. Referring to FIG. 15, a water electrolysis device **1000** may include a plurality of separators **301** and **302** and an electrochemical cell **310** disposed therebetween. The water electrolysis device **1000** may be supplied with water in a form of water vapor as fuel, and the water may be separated into hydrogen and oxygen by electrolysis. To improve a function of the water electrolysis device **1000**, a plurality of units **300** including the separators **301** and **302** and the electrochemical cell **310** may be provided and repeated to form a stack **400**. Referring to components of the units **300**, the units **300** may include a fuel electrode-side separator **301** and an air electrode-side separator **302**. In this case, the fuel electrode-side separator **301** may include a serpentine-type flow path **120** and an auxiliary flow path **130**, connected to each other, as described above, and an inlet I and an outlet O of a second serpentine-type flow path **120b** may be implemented to be disposed closer to a fluid outlet **112** in the first direction D1 than an inlet I and an outlet O of a first serpentine-type flow path **120a**, to improve performance of the water electrolysis device **1000**. In FIG. 15, a separator according to the embodiment of FIG. 1 may be used, but a separator according to another embodiment, such as the separator illustrated in FIG. 4, may also be employed. In addition, a shape of the air electrode-side separator **302** is not limited, and a separator, conventionally used, having a straight wall or the like may be used. The air electrode-side separator **302** may also include a flow path having the same shape as the fuel electrode-side separator **301**.

[0049] For the electrochemical cell **310**, as an example, a solid oxide cell may be used. Specifically, the electrochemical cell **310** may include a fuel electrode **311**, an air electrode **312**, and an electrolyte **313** disposed therebetween. In this case, the fuel electrode **311** may include a cermet layer including a metal-containing phase and a ceramic phase. In this case, the metal-containing phase may include a metal catalyst such as nickel (Ni), cobalt (Co), copper (Cu), alloys thereof, or the like, which acts as an electronic conductor. The metal catalyst may be in a metallic state or may be in an oxide state. The ceramic phase of the fuel electrode **311** may include gadolinia doping seria (GDC), samaria doping seria (SDC), ytterbia doping seria (YDC), scandia stabilized zirconia (SSZ), ytterbia ceria scandia stabilized zirconia (YbCSSZ), or the like. The air electrode **312** may include an electrically conductive material, like an electrically conductive perovskite material such as lanthanum strontium manganite (LSM). Other conducting perovskites, metals such as lanthanum strontium cobalt (LSC), lanthanum strontium cobalt manganese (LSCM), lanthanum strontium cobalt ferrite (LSCF), lanthanum strontium ferrite (LSF), La.sub.0.85Sr.sub.0.15Cr.sub.0.9Ni.sub.0.1O.sub.3 (LSCN) or Pt may also be used. In some embodiments, the air electrode **312** may include a mixture of an electrically conductive material and an ionically conductive ceramic material. For example, the air electrode **312** may include about 10 wt % to about 90 wt % of the electrically conductive material (e.g., LSM or the like) and about 10 wt % to about 90 wt % of the ionically conductive material. In this case, the ionically conductive material may include a zirconia-based material and/or a ceria-based material. The electrolyte **313** may include stabilized zirconia. Specifically, the electrolyte **313** may include scandia stabilized zirconia (SSZ), yttria stabilized zirconia (YSZ), scandia ceria stabilized zirconia (SCSZ), scandia ceria yttria stabilized zirconia (SCYSZ), scandia ceria ytterbia stabilized zirconia (SCYbSZ), or the like.

[0050] Although the electrochemical cell **310** is illustrated above as a solid oxide cell, the electrochemical cell **310** may also be employed as a polymer electrolyte membrane cell or the like.

[0051] Gaskets **321** and **322** may be disposed on an outside of the electrochemical cell **310** between the separators **301** and **302** to prevent fluid from leaking externally. Additionally, current collector layers **323** and **324** may be disposed between the separators **301** and **302** and the electrochemical cell **310**. The current collector layers **323** and **324** may have excellent oxidation resistance to maintain excellent electrical conductivity. Additionally, as illustrated, the current collector layers **323** and **324** may have a porous structure such as a net structure to allow fluid to pass therethrough.

[0052] In a separator for an electrochemical device according to an example of the present disclosure, high performance may be provided when applied to the electrochemical device.

[0053] While example embodiments have been illustrated and described above, it will be apparent to those skilled in the art that modifications and variations could be made without departing from the scope of the present disclosure as defined by the appended claims.

Claims

1. A separator for an electrochemical device, comprising: a fluid inlet; a fluid outlet; a plurality of serpentine-type flow paths disposed between the fluid inlet and the fluid outlet; and an auxiliary flow path extending in a first direction and connected to inlets and outlets of the plurality of serpentine-type flow paths, the first direction being a direction from the fluid inlet toward the fluid outlet, wherein the plurality of serpentine-type flow paths include a first serpentine-type flow path and a second serpentine-type flow path, and an inlet and an outlet of the second serpentine-type flow path are disposed closer to the fluid outlet in the first direction, compared to an inlet and an outlet of the first serpentine-type flow path.
2. The separator of claim 1, wherein the auxiliary flow path is a straight flow path.
3. The separator of claim 1, wherein the auxiliary flow path is parallel to the first direction.
4. The separator of claim 1, wherein the auxiliary flow path is connected to the fluid inlet and the fluid outlet, to have an integral structure.
5. The separator of claim 1, wherein the first serpentine-type flow path is provided as a pair on one side and the other side of the auxiliary flow path.
6. The separator of claim 1, wherein an inlet of a serpentine-type flow path disposed closest to the fluid inlet, among the plurality of serpentine-type flow paths, is connected to the fluid inlet.
7. The separator of claim 1, wherein an outlet of a serpentine-type flow path disposed closest to the fluid outlet, among the plurality of serpentine-type flow paths, is connected to the fluid outlet.
8. The separator of claim 1, wherein the auxiliary flow path is disposed in a central region in a second direction perpendicular to the first direction.
9. The separator of claim 8, having a symmetrical structure, based on the auxiliary flow path.
10. The separator of claim 1, wherein the plurality of serpentine-type flow paths comprise a first flow path in the first direction and a second flow path in a second direction perpendicular to the first direction.
11. The separator of claim 10, wherein the second flow path is provided as a plurality of second flow paths, the first flow path is connected to the plurality of second flow paths, and the plurality of second flow paths are connected to the auxiliary flow path.
12. The separator of claim 1, wherein the auxiliary flow path is provided as a plurality of auxiliary flow paths, and the fluid inlet and the fluid outlet are provided as a plurality of fluid inlets and as a plurality of fluid outlets, respectively, to be connected to the auxiliary flow path.
13. The separator of claim 12, wherein the plurality of auxiliary flow paths are disposed in an outer peripheral region in a second direction perpendicular to the first direction.
14. The separator of claim 13, wherein the plurality of serpentine-type flow paths comprise a first flow path in the first direction and a plurality of second flow paths in the second direction, the first flow path is connected to the plurality of second flow paths, and the plurality of second flow paths

are connected to the plurality of auxiliary flow paths.

15. The separator of claim 14, wherein the first flow path is disposed in a central region in the second direction.

16. The separator of claim 15, having a symmetrical structure, based on the first flow path.

17. The separator of claim 1, wherein a width, in a second direction perpendicular to the first direction, of the auxiliary flow path is less than a width, in the first direction, of each of the first serpentine-type flow path and the second serpentine-type flow path.

18. An electrochemical device comprising: a plurality of separators; and an electrochemical cell disposed between the plurality of separators, wherein at least one of the plurality of separators includes a fluid inlet; a fluid outlet; a plurality of serpentine-type flow paths disposed between the fluid inlet and the fluid outlet; and an auxiliary flow path extending in a first direction and connected to inlets and outlets of the plurality of serpentine-type flow paths, the first direction being a direction from the fluid inlet toward the fluid outlet, and wherein the plurality of serpentine-type flow paths include a first serpentine-type flow path and a second serpentine-type flow path, and an inlet and an outlet of the second serpentine-type flow path are disposed closer to the fluid outlet in the first direction, compared to an inlet and an outlet of the first serpentine-type flow path.
